

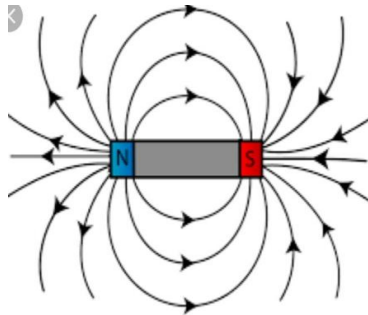
Module IV

MAGNETISM AND ELECTROMAGNETIC THEORY

MAGNETIC FIELD

It is the area around a magnet in which there is magnetic force. It is a vector quantity

- Represented either by 'B' or 'H'



MAGNETIC FIELD INTENSITY (Magnetising Field) (H)

The magnetic field intensity, H is defined as the magnetic force experienced by a unit positive charge in motion. It is a vector quantity

MAGNETIC FLUX DENSITY (B)

The magnetic flux density is defined as the total Number of magnetic field lines passing perpendicularly through unit area. It is expressed Weber/sq.meter Or Tesla . If ϕ is the magnetic flux and A is the area, then magnetic flux density

$$B = \frac{\phi}{A}$$

Magnetic flux density (B) and magnetic field intensity (H) are related by

$$B = \mu H$$

Where μ is the permeability of the medium

GAUSS'S LAW FOR MAGNETIC FLUX

Magnetic flux through a area is given by

$$\phi = \int_S \vec{B} \cdot d\vec{s}$$

The lines of B are continuous. All magnetic fields originate from current loops. Magnetic lines around such loops are always closed. Therefore, the total magnetic flux through a closed surface is zero.

By Gauss's Law, the surface integral of magnetic flux over a closed surface is zero.

$$\int_S \vec{B} \cdot d\vec{S} = 0$$

Physical significance

- The number of magnetic field lines entering in to a closed surface is equal to the number of lines passing out of the surface.
- Magnetic monopole does not exist.
- There are no magnetic charges and therefore no start or end for magnetic flux.

AMPERE'S CIRCUITAL LAW

Oersted observed that a magnetic field is always produced around a conductor carrying current. Ampere's circuital law explains the relationship between the current and the magnetic field created by it.

It states that the line integral of the magnetic flux density (B) for a closed path is equal to μ_0 times the net current I enclosed by the path.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

Proof

Consider a long straight conductor carrying current I. Magnetic lines of force are concentric circles in a plane normal to the length of the conductor.

The magnitude of magnetic field due to a long straight conductor is given by

$$B = \frac{\mu_0 I}{2\pi r}$$

we can see in the diagram that angle between the element $d\vec{l}$ and magnetic field $\vec{B}$ is 0°

$$\int \vec{B} \cdot d\vec{l} = \int B dl \cos 0 = B \int dl$$

Also, the integral of element will form the whole circle of circumference ($2\pi r$):

$$\int dl = 2\pi r$$

Now putting the value of B and $\int dl$ in the equation, we get:

$$B \int dl = \frac{\mu_0 I}{2\pi r} \times 2\pi r = \mu_0 I$$

$\therefore$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

Whenever the magnetic flux linked with a circuit changes an emf is induced in that circuit.

According to Faradays Law ,The magnitude of induced emf is equal to the rate of change of magnetic flux through the circuit.

$$\text{Induced emf, } e = \frac{d\Phi}{dt}$$

$$e = - \frac{d\Phi}{dt} \text{ (Lenz's law)}$$

-ve sign indicates the direction of the induced emf which tends to oppose the change in the magnetic flux.

INTENSITY OF MAGNETISATION (M)

It is a measure of the magnetisation of a magnetised specimen.

It is defined as the magnetic moment per unit volume of the material.

$$M = \frac{\text{Magnetic Moment}}{\text{Volume}}$$

$$\text{But Moment} = m * 2l$$

$$M = \frac{m * 2l}{V}$$

M is pole strength, 2l the total length of specimen.

It is measured in ampere per meter.

MAGNETIC SUSCEPTIBILITY

The magnetic susceptibility is a measure of how much a material become magnetised in an applied magnetic field. It is defined as the ratio of magnetization M to the applied magnetizing field intensity H.

$$\chi = \frac{M}{H}$$

Since M and H are measured in amper per meter is dimensionless quantity

MAGNETIC PERMEABILITY

The magnetic permeability is defined as the property of the material to allow the magnetic lines of force to pass through it.

The permeability of the material is equal to the ratio of the flux density B in the medium to the field intensity H.

$$\mu = \frac{B}{H}$$

RELATIVE PERMEABILITY

The relative permeability of the material is the comparison of the permeability with the permeability of air or vacuum

Relative permeability

$$\mu_r = \frac{\mu}{\mu_0}$$

μ_0 is the permeability of free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ Henry /meter}$$

When a magnetic material is placed in a magnetic field of uniform intensity H, magnetic flux pass through it due to the magnetized field H and due to the material being magnetized by induction(M).

The flux density B in the magnet is

$$B = \mu_0(H + M)$$

$$B = \mu_0 H + \mu_0 M$$

$$\frac{B}{H} = \frac{\mu_0 H}{H} + \frac{\mu_0 M}{H}$$

$$\mu = \mu_0 + \mu_0 \chi$$

$$\mu = \mu_0(1 + \chi)$$

$$\frac{\mu}{\mu_0} = (1 + \chi)$$

$$\mu_r = 1 + \chi$$

CLASSIFICATION OF MAGNETIC MATERIALS

Materials that can be magnetized in the presence of magnetic field are known as magnetic materials.

Magnetic properties are induced in the material in presence of external magnetic field. On the basis of different magnetic properties are classified in to different types.

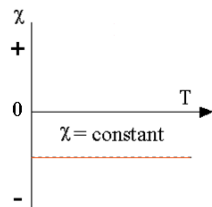
1. Diamagnetism
2. Para magnetism
3. Ferro magnetism

DIAMAGNETISM

- Diamagnetic materials are substances which have tendency to move from stronger to weaker magnetic field when placed in non uniform magnetic field.
- Repelled by the applied magnetic field. (weak repulsion)
- If a rod of diamagnetic material is suspended in uniform field it will orient perpendicular to the field
- Material having even atomic number
- Atomic shells are filled and have no unpaired electron
- Magnetic moments are equal and opposite for a pair of electrons
- The atoms or molecules possess zero magnetic moment.

When magnetic field is applied,

- Electrons having orbital moment in the same direction of applied Magnetic Field slow down and those in the opposite direction speed up.
- The orbital motion gets distorted.
- Net magnetic moment is produced which opposes the applied field.
- Result in weak repulsion
- Posses' very weak susceptibility and is negative.
- χ is independent of temperature.
- Permeability is less than one
- Examples are Copper, Bismuth, Lead, water, cadmium, zinc, helium.



Langevin theory

Langevin theoretically calculated the susceptibility of a diamagnetic material, taking the magnetic moment produced due to orbital motion of electron.

$$\text{Susceptibility, } \chi = -\frac{\mu_0 e^2 n z}{6m} \langle r^2 \rangle$$

μ_0 - permeability of free space, n - No of atoms per unit volume,

z - Atomic number, e - Electronic charge, m - Mass of electrons

$\langle r^2 \rangle$ - Mean square distance of electron from the nucleus

PARAMAGNETISM

- Paramagnetic materials are weakly magnetised in an externally applied magnetic field.
- Paramagnetic material show tendency to move from weak to strong magnetic field when placed in a non uniform magnetic field.
- Paramagnetism is due to the presence of unpaired electrons in atoms.

- Electrons undergo spin motion and orbital motion and produce spin magnetic moment and orbital magnetic moment.
- These moments are randomly oriented and exhibit no magnetic moment

When magnetic field is applied,

- The electron's spin tends to align with the applied field resulting in magnetic moment
- Induced field will be in the direction of applied field; but weak (weak attraction)
- Free electron magnetism is weak and is called Pauli Paramagnetism
- They do not retain any magnetism in the absence of an external magnetic field.
- Susceptibility is low and positive.
- Paramagnetic susceptibility depends on temperature
- Examples: aluminium, chromium, calcium, platinum, O_2 , Ne,

CURIE'S LAW

According to Curie's law, the susceptibility of a paramagnetic material is inversely proportional to its temperature.

$$\chi \propto \frac{1}{T}$$

$$\chi = \frac{C}{T}$$

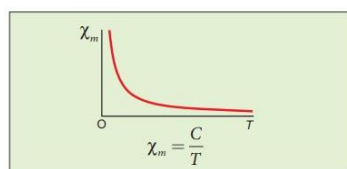
C is called Curie's constant and is given by

$$C = \frac{\mu_0 n \mu^2}{3k}$$

μ_0 = Permeability of free space, n = no. of atoms per unit volume.

μ = total magnetic moment due to spin and orbital motion.

k = Boltzmann constant.



FERROMAGNETISM

- Their magnetic property not only because of their atomic magnetic moment but also due to certain region called magnetic domain
- In a domain very large number of atoms are aligned parallel to each other.
- Magnetic force within the domain is very strong.
- Before magnetisation the domains are randomly oriented relative to each other (net dipole moment zero).

When magnetic field is applied,

- Domains within the material are aligned
- When all domains are aligned the material is said to be magnetically saturated
- Strongly attracted by external field
- Magnetism persists even after the removal of magnetic field.
- They show hysteresis.
- Loss their magnetic properties if they are heated to high temperatures.
- They are used to make permanent magnet.
- They have high susceptibility and relatively high permeability.
- Examples: iron, nickel, cobalt, magnetite

CURIE-WEISS LAW

According to Curie Weiss Law, at above Curie temperature the magnetic susceptibility of a ferromagnetic material is inversely proportional to $(T - T_c)$

$$\chi = \frac{C}{T - T_c}$$

where C is Curie's constant, T_c is the transition temperature or Curie's temperature.

Curie temperature is the temperature above which a Ferromagnetic material changes to paramagnetic material.

WEISS THEORY

The magnetic moment of atoms associated with ferro magnetic substance are of spin origin. Ferromagnetic specimen contains a number of small regions (domain) .

Weiss predicted that, in ferromagnetic materials spontaneous magnetization is observed, which is due to a strong internal field arising from an exchange interaction between the magnetic moments in the neighbourhood domains

The magnitude of spontaneous magnetisation is the sum of magnetic moments of the individual domains.

The internal field is equal to intermolecular field of strength H_m which is proportional to the magnetization M within the domain

$$H_m \propto M$$

$$H_m = \lambda M$$

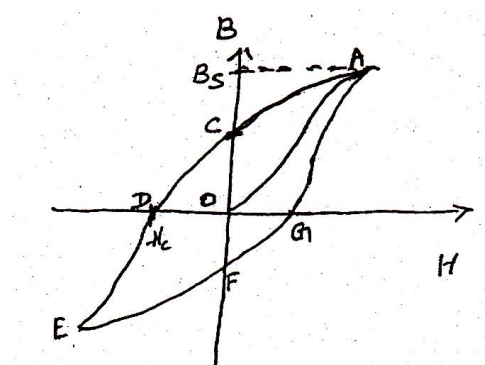
where λ is constant called the Weiss constant.

EXCHANGE INTERACTION THEORY

- In Ferro magnetic materials like Fe, Ni, Co the 3d orbitals are partially filled.
- There are 4 unpaired electrons in Fe.
- Incomplete 3d orbital electron interacts strongly with neighbour. Result is zero magnetic moment to orbital motion. But magnetic moment due to spin motion exists.
- If the electrons interact very strongly then their spins will be aligned in the same direction. But it is short range interaction.
- Each in changes the magnetic moment of its neighbouring ion.
- This makes the alignment of all the magnetic moment in the material producing magnetisation

HYSTERESIS CURVE

Consider a completely demagnetized ferromagnetic material. It is subjected to increasing values of magnetic field intensity H and the corresponding flux density B is measured. The variation of B with H is shown in the figure.



When we increase the magnetic field intensity H beyond A, the flux density, B will not change any more. In this condition the material is said to be saturated. And

the B value is called saturated flux density. When the intensity H is decreases, the flux density B will not retrace the curve OA. It will follow the curve AC. If the field intensity H is reduced to zero, some flux remains in the material .This magnetization is called remanent flux density.

In order to reduce B value to zero, a magnetic field is applied in the opposite direction. The negative field H required to make B to zero is called coercive field.

Here the flux density changes lags behind the changes in the magnetic field .This is called Hysterisis. Curve is called Hysteresis loop

- Remanent flux density- B_r In the figure it is OC
- Coercive Field- H_c In the figure it is Od

Electromagnetic theory

Clark Maxwell proposed electromagnetic theory which united all the phenomena of electricity, magnetism and light

Del Operator

$$\nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

del operator = differential operator

Where i, j, k are the unit vectors along the Cartesian coordinate axes

Gradient of a Scalar function

If the differential operator is operated on a scalar function f, the operation is known as gradient of a scalar function.

$$\text{Grad } f = \nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

Grad f is a vector quantity, even though the operational function is scalar function

Physical significance

- If the scalar function f represents temperature, the ∇f represents the rate of change of temperature with distance or the temperature gradient.
- Temperature is a scalar quantity but temperature gradient is a vector quantity having both magnitude and direction
- The direction of the resultant and is the direction in which the temperature gradient is maximum

Divergence of a vector

It is Multiplication of ∇ operator by another vector and the result is a scalar product. This operation is called divergence of a vector.

If the vector A is represented by

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

Divergence A or Div A = $\nabla \cdot A$

$$\begin{aligned} &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \\ &= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \end{aligned}$$

Physical significance

- At any point if the Outflow is greater than inflow, divergence is said to be +ve at that point
- If the Outflow is less than inflow, divergence is said to be -ve
- For an incompressible fluid, like water, Outflow is equal to inflow and the divergence is zero.
- So the divergence of the fluid at any point gives the idea whether the fluid is compressible or not.

Curl of a vector

Another operation with the del operator is the cross product with another vector. It is called curl of a vector and is defined by

$$\text{Curl } \vec{A} = \nabla \times \vec{A}$$

$$\nabla \times \vec{A} =$$

$$= \hat{i} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) - \hat{j} \left(\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right) + \hat{k} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

Curl of a vector is also a vector

Physical significance

- The curl of a vector field measures the tendency of the Vector field to rotate about a point.
- Curl of a vector field at a point is a Vector that points in the Direction of the axis of rotation and has magnitude represents speed of rotation
- In stream flow problems, the concept of curl or rotation of vector quantity is used
- curl $\vec{v}$ is not zero in stream motion
- Velocity field of water flowing out of Sink acquires a circulation, because the curl of a velocity vector field is not zero.
- curl of a vector field is not zero - rotational
- curl of a vector field is zero – irrotational
- It is a measure of how much the vector 'curls around' or rotate around the point.
- In Hydrodynamics and aerodynamics, this concept has importance

GAUSS THEOREM(Relation between surface integral and volume integral)

Gauss theorem relates the surface integral and volume integral . For a closed surface s , surrounding the volume V , Gauss theorem states that the volume integral of divergence of $\vec{A}$ is equal to the surface integral of $\vec{A}$ over the closed surface.

$$\int_s \vec{A} \cdot d\vec{s} = \int_v (\nabla \cdot \vec{A}) dv$$

STOKE'S THEOREM

Stokes theorem relates the line integral and surface integral. If S is a surface bounded by a closed curve C , then Stokes theorem states that the surface integral of curl A over a surface is equal to line integral of A along the closed path around the surface

$$\int_l \vec{A} \cdot d\vec{l} = \int_s (\nabla \times \vec{A}) \cdot d\vec{s}$$

EQUATION OF CONTINUITY

The total current flowing out of some volume must be equal to the rate of decrease of charge within the volume, since the charge is conserved.

In mathematical terms, this conservation of charge concept can be stated as

$$I = - \frac{\partial q}{\partial t}$$

$$\int J \cdot d\vec{s} = - \frac{\partial}{\partial t} \int \rho \, dv$$

Where ρ is the charge per unit volume or volume charge density. ($\rho = q/v$)

$$\int J \cdot d\vec{s} = - \int \frac{\partial}{\partial t} \rho \, dv$$

By using Gauss theorem, convert surface integral to volume integral

$$\int (\nabla \cdot J) \, dv = - \int \frac{\partial}{\partial t} \rho \, dv$$

To hold this equation the integrands must be equal

$$\nabla \cdot J = \frac{-\partial \rho}{\partial t} \quad \nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

This is the equation of continuity for time varying fields.

MAXWELL'S EQUATIONS

Maxwell derived four relations to describe the motion of electromagnetic waves in different media. These are known as Maxwell's equations

1. $\nabla \cdot D = \rho$

2. $\nabla \cdot B = 0$

3. $\nabla \times E = - \frac{\partial B}{\partial t}$

4. $\nabla \times H = J + \frac{\partial D}{\partial t}$

MAXWELL'S FIRST EQUATIONS

Maxwell's First equation is the statement of Gauss's Law in electrostatics, it is also called Electric flux equation

According to Gauss's law in electrostatics, The total normal electric flux through a closed surface is equal to $\frac{1}{\epsilon_0}$ times the total charge enclosed by the surface.

$$\text{That is } \phi_E = \frac{q}{\epsilon_0}$$

$$\int_s E \cdot ds = \frac{q}{\epsilon_0}$$

$$q = \int_v \rho dv$$

$$\int_s E \cdot ds = \frac{1}{\epsilon_0} \int_v \rho dv$$

$$\int \epsilon_0 E \cdot ds = \int \rho dv$$

$D = \epsilon_0 E$ is called electric displacement vector

$$\int_s D \cdot ds = \int_v \rho dv$$

Applying Gauss theorem

$$\int_v \nabla \cdot D dv = \int_v \rho dv$$

To hold this equation the integrands must be equal.

$\nabla \cdot D = \rho$Maxwell's First equation

MAXWELL'S SECOND EQUATIONS

Second equation explains Gauss Law in Magnetism

By Gauss's law in magnetism the net magnetic flux emerging through any closed surface is zero.

$$\text{Then } \int_s B \cdot ds = 0$$

By using Gauss theorem,

$$\int_v \nabla \cdot B \, dv = 0$$

$$\nabla \cdot B = 0 \dots \dots \dots \text{Maxwell's second equation}$$

For a magnetic dipole, in any closed surface the magnetic flux directed inwards towards South Pole will be equal to the flux outward from North Pole. The net flux will be zero for a dipole.

MAXWELL'S THIRD EQUATIONS

Third equation describes Faraday's law of electromagnetic Induction

According to Faradays law of electromagnetic Induction,

$$\text{Induced emf } e = -\frac{\partial \phi}{\partial t}$$

ϕ is magnetic flux

$$e = -\frac{\partial}{\partial t} \int B \cdot ds \dots \dots \dots (1)$$

$$\text{We know that } e = \int E \cdot dl \dots \dots \dots (2)$$

$$\int_l E \cdot dl = -\frac{\partial}{\partial t} \int_s B \cdot ds$$

Using Stokes theorem

$$\int_s \nabla \times E \cdot ds = -\int_s \frac{\partial B}{\partial t} \cdot ds$$

To hold this equation Integrands must be equal

$$\nabla \times E = -\frac{\partial B}{\partial t} \dots \dots \dots \text{Maxwell's third equation}$$

MAXWELL'S FOURTH EQUATIONS

Fourth equation is the statement of modified Ampere's circuital law and is the equation for induced magnetic field

According to amperes circuital law the magnetic induction around a closed current carrying loop is equal to μ_0 times the current I flowing in the loop.

$$\text{Thus } \int B \cdot dl = \mu_0 I$$

$$\int \mu_0 H \cdot dl = \mu_0 I$$

$$\int H \cdot dl = I \dots\dots\dots(1)$$

If J is the current density of the loop,

then current

$$I = \int J \cdot ds \dots\dots\dots(2)$$

From eqn 1 and 2

$$\int_l H \cdot dl = \int_s J \cdot ds$$

Using Stokes theorem

$$\int_s \nabla \times H \cdot ds = \int_s J \cdot ds$$

$$\nabla \times H = J \dots\dots\dots(4)$$

From eqn of continuity

$$\nabla \cdot J = -\frac{\partial \rho}{\partial t} \dots\dots\dots(5)$$

Take divergent of eqn (4)

$$\nabla \cdot \nabla \times H = \nabla \cdot J$$

But $\nabla \cdot \nabla \times H = 0$ Since divergent of curl of any vector will be zero.

$$\text{Hence } \nabla \cdot J = 0 \dots\dots\dots(6)$$

Here there is incompatibility in eqn 5 and 6

Hence Maxwell introduced a new term J_D called displacement current density

$$\nabla \times H = J + J_D \dots\dots\dots(7)$$

Now take divergent on both sides of eqn (7)

$$\nabla \cdot \nabla \times H = \nabla \cdot J + \nabla \cdot J_D$$

$$0 = \nabla \cdot J + \nabla \cdot J_D$$

$$\nabla \cdot J = -\nabla \cdot J_D$$

From eqn 5

$$-\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J}_D$$

From Maxwell's first law

$$\rho = \nabla \cdot \mathbf{D}$$

$$\frac{\partial}{\partial t} \nabla \cdot \mathbf{D} = \nabla \cdot \mathbf{J}_D$$

$$\nabla \cdot \frac{\partial \mathbf{D}}{\partial t} = \nabla \cdot \mathbf{J}_D$$

$$\frac{\partial \mathbf{D}}{\partial t} = \mathbf{J}_D$$

now substitute this value in eqn7

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \dots \dots \dots \text{This is Maxwell's fourth equation}$$

SIGNIFICANCE OF $\frac{\partial \mathbf{D}}{\partial t}$

- The term physically indicates that not only a current produces a magnetic field but a changing electric field also produces a magnetic field.
- It is the presence of this term $\frac{\partial \mathbf{D}}{\partial t}$ which leads to the production of em waves.
- A changing Electric field produces a varying magnetic field and changing magnetic field produces a varying Electric field.

CONDUCTION CURRENT

Conduction current is due to the movement of electric charges in a conductor when an electric field is applied. Consider a conductor of length l and of cross sectional area A . I be the current passing through it and V be the potential difference between the ends of the conductor.

$$\text{The resistance } R = \frac{l}{\sigma A}$$

Where σ is the conductivity

$$\text{By Ohms law } V = IR$$

$$\text{Also } V = E l$$

$$IR = E l$$

$$E l = I \frac{l}{\sigma A}$$

$$E = \frac{I}{\sigma A} = \frac{J}{\sigma}$$

$$J = \sigma E$$

J is the conduction current per unit area and is directly proportional to E.

So the conduction current is due to the flow of electric charges in a conductor.

DISPLACEMENT CURRENT

It is the current produced due to the rate of change of electric flux

Consider a parallel plate capacitor of capacitance C.

$$C = \frac{\epsilon_0 A}{d} \quad d \text{ is the separation between the plates.}$$

$$Q = CV$$

The electric field

$$E = \frac{V}{d} = \frac{Q}{Cd} = \frac{Q}{\epsilon_0 A}$$

Electric flux

$$\phi_E = EA = \frac{QA}{\epsilon_0 A} = \frac{Q}{\epsilon_0}$$

Then Current

$$I = \frac{dQ}{dt}$$

$$Q = \epsilon_0 \phi_E$$

$$\frac{dQ}{dt} = \epsilon_0 \frac{d\phi}{dt}$$

Here $\frac{dQ}{dt}$ is the current I produced due to the rate of change of electric flux. Thus a changing E.F produces a current and it is called displacement current.

Difference between conduction current and displacement current

- Conduction current is produced due to rate of change of charges. But the displacement current is produced due to rate of change of Electric Field.
- Conduction current is zero in perfect vacuum but displacement current has a finite value in vacuum.
- Conduction current leads the EF where as Displacement current leads the Electric Field where as the Displacement current leads the electric field by $\pi/2$ radians

ELECTROMAGNETIC WAVES

A varying Electric field produces a varying magnetic field and varying magnetic field produces a varying Electric field. Thus E and B are closely related. When the time varying Electric field give rise to time varying Magnetic field, this varying magnetic field again produces varying electric field. And this process continues. Thus E and B are coupled and propagate in space as Electromagnetic wave. and travels with velocity of light. EM waves are transverse in nature.

PROPAGATION OF ELECTROMAGNETIC WAVES IN FREE SPACE

$$\nabla \cdot D = \rho$$

$$\nabla \cdot B = 0$$

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\nabla \times H = J + \frac{\partial D}{\partial t}$$

In vacuum, $J=0$, $\rho=0$

Hence the Maxwells equations becomes

$$\nabla \cdot D = 0$$

$$\nabla \cdot B = 0$$

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\nabla \times H = \frac{\partial D}{\partial t}$$

Again in free space $D = \epsilon_0 E$, $B = \mu_0 H$

$$\nabla \cdot E = 0 \dots\dots\dots(1)$$

$$\nabla \cdot H = 0 \dots\dots\dots(2)$$

$$\nabla \times E = -\mu_0 \frac{\partial H}{\partial t} \dots\dots\dots(3)$$

$$\nabla \times H = \epsilon_0 \frac{\partial E}{\partial t} \dots\dots\dots(4)$$

Taking curl on both sides of eqn (3)

$$\nabla \times \nabla \times E = -\mu_0 \nabla \times \frac{\partial H}{\partial t} \dots\dots\dots(5)$$

In vector Calculus

$$\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$$

Eqn (5) becomes

$$\begin{aligned} \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} &= -\mu_0 \nabla \times \frac{\partial \mathbf{H}}{\partial t} \\ -\nabla^2 \mathbf{E} &= -\mu_0 \frac{\partial}{\partial t} \nabla \times \mathbf{H} & (\nabla \cdot \mathbf{E} = 0) \\ \nabla^2 \mathbf{E} &= \mu_0 \frac{\partial}{\partial t} \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \\ \nabla^2 \mathbf{E} &= \mu_0 \epsilon_0 \frac{\partial}{\partial t} \frac{\partial \mathbf{E}}{\partial t} \\ \nabla^2 \mathbf{E} &= \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \dots \dots \dots (6) \end{aligned}$$

Taking curl on both sides of eqn (4)

$$\begin{aligned} \nabla \times \nabla \times \mathbf{H} &= \epsilon_0 \nabla \times \frac{\partial \mathbf{E}}{\partial t} \\ \nabla(\nabla \cdot \mathbf{H}) - \nabla^2 \mathbf{H} &= \epsilon_0 \frac{\partial}{\partial t} \nabla \times \mathbf{E} & (\nabla \cdot \mathbf{H} = 0) \\ -\nabla^2 \mathbf{H} &= -\epsilon_0 \mu_0 \frac{\partial}{\partial t} \frac{\partial \mathbf{H}}{\partial t} \\ \nabla^2 \mathbf{H} &= \epsilon_0 \mu_0 \frac{\partial^2 \mathbf{H}}{\partial t^2} \dots \dots \dots (7) \end{aligned}$$

Comparing equations 6 and 7 with the standard eqn of wave motion

$$\begin{aligned} \nabla^2 \psi &= \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \\ \frac{1}{v^2} &= \epsilon_0 \mu_0 \\ v &= \frac{1}{\sqrt{\epsilon_0 \mu_0}} \\ v &= 3 \times 10^8 \text{ m/s} \end{aligned}$$

Thus electromagnetic waves travel in freespace with the velocity of light.

ENERGY OF ELECTROMAGNETIC WAVES

When EM waves travels in free space or vacuum the energy density u_e in the electric field and u_m in magnetic field is given by

$$\begin{aligned} u_e &= \frac{1}{2} \epsilon_0 E^2 \\ u_m &= \frac{1}{2} \mu_0 H^2 \end{aligned}$$

POYNTING VECTOR

The magnitude and direction of flow of energy per unit area per unit time in an Electromagnetic wave travelling in freespace is represented by a vector known as Poynting vector.

$$S = E \times H$$

unit is Watt per square meter.

Poynting vector represents the instantaneous power density associated with the EM field at a given point.

$$S = \frac{P}{A}$$

